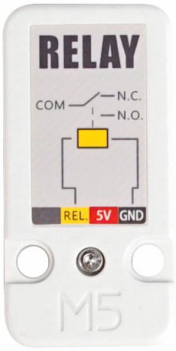
Unit Relay

SKU:U023













Description

Unit Relay is a single-channel relay control high-voltage driving unit. It is based on the HK4100F-DC5V-SHG relay and provides SPDT (Single Pole Double Throw) switching, supporting high-voltage on/off control up to AC 220V@3A / DC 30V. This unit is directly driven by a single GPIO logic level and features physical isolation as well as a status indicator LED. It communicates via a Grove HY2.0-4P interface and adopts a LEGO-compatible hole design, enabling flexible integration with LEGO structures or screw mounting. It is suitable for safe power management scenarios in smart homes, appliance control, and automation equipment.

Features

- Channel Type: 1 x SPDT Relay
- Load Capacity: AC 220V@3A or DC 30V@3A
- Status Indicator: Relay Status LED
- Control Method: Direct Single GPIO Level Drive
- Communication Interface: Grove HY2.0-4P
- 2 x LEGO-Compatible Holes
- Development Platform
  - UiFlow1
  - UiFlow2
  - Arduino IDE

Includes

- 1 x Unit Relay
- 1 x HY2.0-4P Grove cable (20cm)
- 1 x 3.96 terminal

Applications

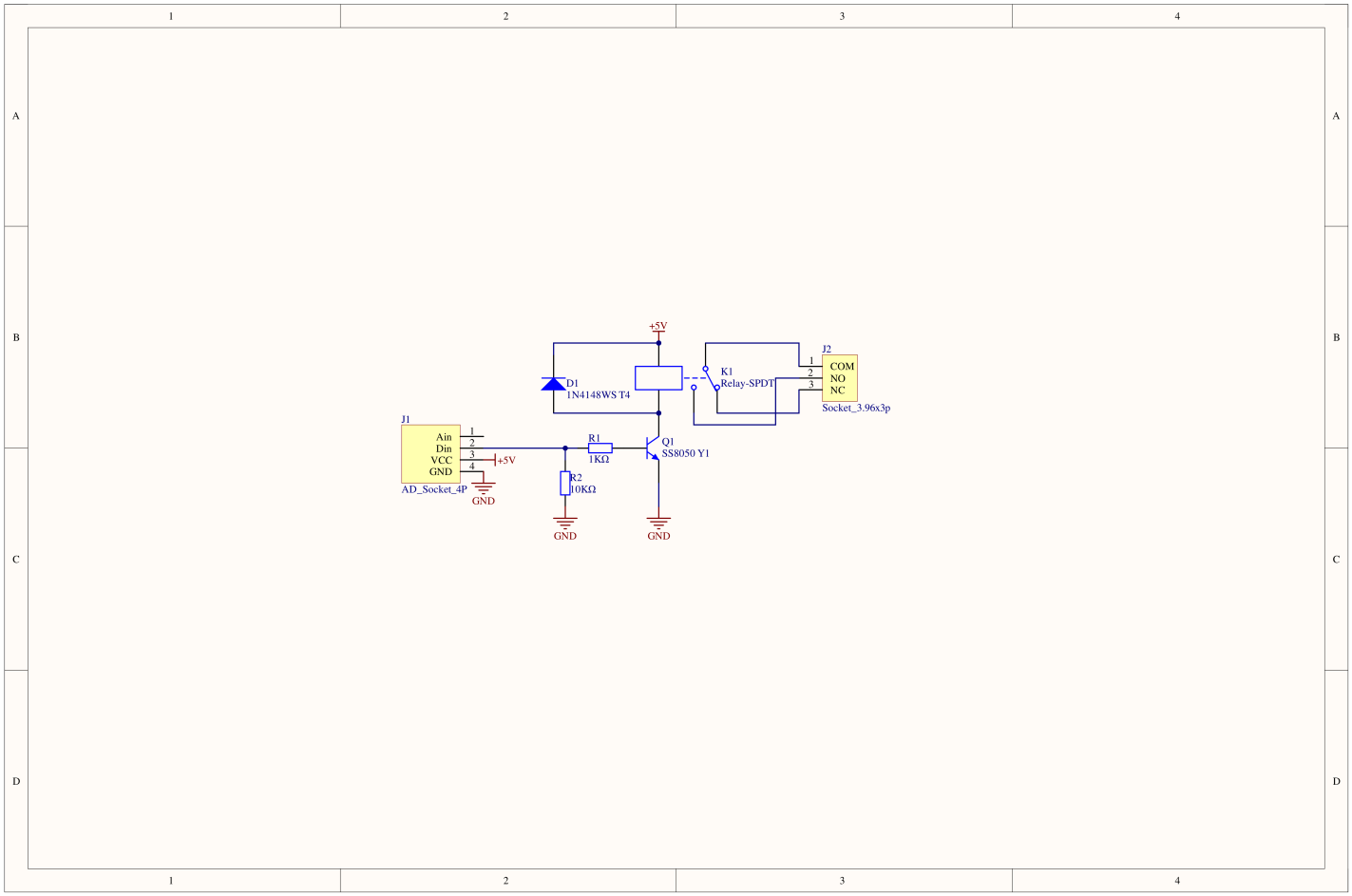
- Remote control of high-power appliances such as refrigerators, air conditioners, TVs, etc.

Specifications

Specification	Parameter
Single relay power consumption	5V@40mA
Coil action time	6ms
Coil release time	4ms
Product Size	48.0 x 24.0 x 8.0mm
Product Weight	9.5g
Package Size	138.0 x 93.0 x 9.0mm
Gross Weight	17.1g

Schematics

- Unit Relay Schematics PDF

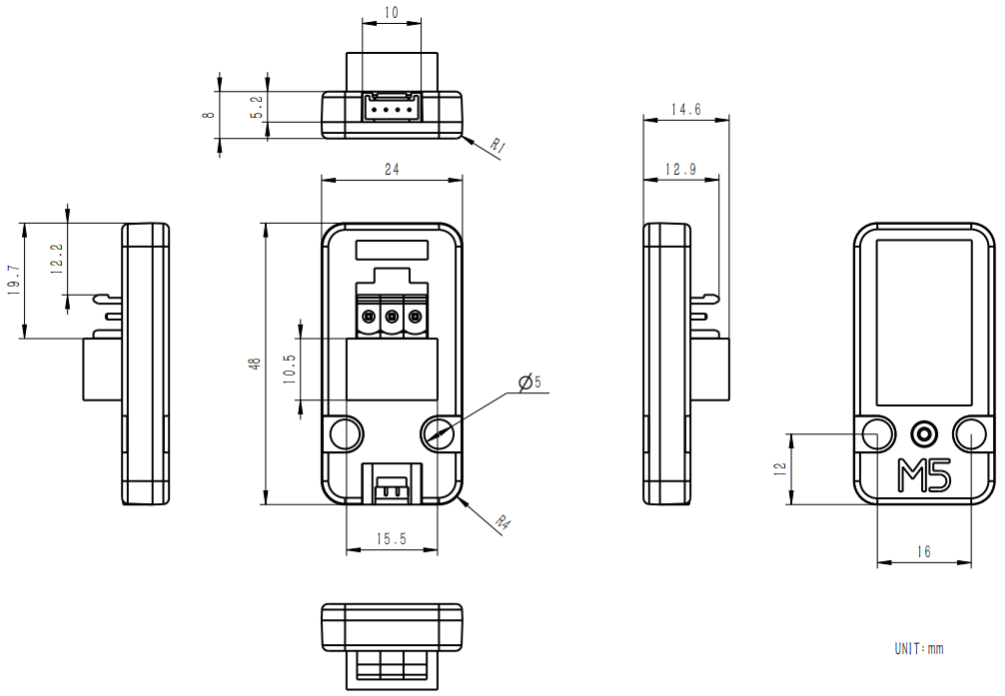


PinMap

Unit Relay

HY2.0-4P	Black	Red	Yellow	White
PORT.B	GND	5V	DIN	NC

Model Size



Structure

- Unit Relay Structure Files

Datasheets

- Relay Datasheet

Softwares

Arduino

- Unit Relay Arduino Tutorial

UiFlow2

- UiFlow2 Docs

EasyLoader

Easyloader	Download Link	Notes
Unit Relay Test Easyloader	download	/

Video

- Relay coil cycling on/off, contacts used as switches for circuit control.

Relay_UNIT.mp4

- **Controlling Home Lights with UIFlow and RELAY Unit**

Blinking a bulb with the M5 Relay unit..mp4